CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of MSc Physics Examination March 2018

PS713 Thermodynamics and Statistical Mechanics

Date: 28/03/2018 Day: Wednesday Time: 10:00AM To 10:30AM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. **20**

1. The Sterling's approximation for large value of N is

[A] $\ln N! = N \ln N + N$ [B] $\ln N! = N \ln N - N$ [C] $\ln N! = \ln N - N$

[D] $\ln N! = \ln N + N$

2. The entropy of a system consisting 3 coins is

[A] $\ln 2$ [B] $\ln 3$ [C] $\ln 4$ [D] 2.079 [E] none

3. A system is said to be in thermodynamic equilibrium if the system is in

[A] mechanical and chemical equilibrium [B] thermal equilibrium [C] thermal and mechanical equilibrium [D] [A]+[B]

4. For an ideal gas undergoing adiabatic process which of the following equations hold

[A] $TV^{(\gamma-1)}$=constant [B] TV^{γ}=constant [C] $TV^{(\gamma+1)}$ [D] TV=constant

5. Which of the following is a set of intensive variables?

[A] {pressure, tension in a wire, surface tension}
[B] {pressure, chemical potential, magnetization}
[C] {Electric field, polarization, magnetic field}
[D] {chemical potential, charge, Length}

6. Which one of the following is not a measurable quantity?

[A] Heat [B] temperature [C] mass [D] Volume

7. For an ideal gas the volume expansion coefficient is equal to

[A] T [B] $1/T$ [C] $2T$ [D]$1/T^2$

8. A sample of ideal gas has an internal energy U and is then compressed to one half of its original volume while the temperature stays the same. What is the new internal energy of the ideal gas in term of U?

[A] U [B] (1/2)U [C] (1/4)U [D] 2U

9. The integration of a state function over a cyclic process is

[A] one [B] zero [C] not defined [D] infinity

10. The ratio of specific heats $\gamma (= C_P/C_V)$ for a diatomic ideal gas is

[A] 7/5 [B] 5/3 [C] 2 [D] 2/3

11. The Boltzman equation of entropy is

[A] $S = k \ln \Omega$ [B] $S = \ln \Omega$ [C] dQ/T [D] $S = kT \ln \Omega$ [E] None

12. The canonical partition function of a system having discrete energy levels with degeneracy g_i of energy level E_i is given by

[A] $\sum_i g_i \exp(-\beta E_i)$ [B] $\sum_i \exp(-\beta g_i E_i)$ [C] $\sum_i \exp(-\beta E_i)$ [D] $\sum_i \frac{\exp(-\beta E_i)}{g_i}$

13. The stationary ensemble is defined by which of the following?

[A] $\partial\rho/\partial t = H$ [B] $\partial\rho/\partial t = [\rho, H]$ [C] $\partial\rho/\partial t = 0$ [D] $\partial\rho/\partial t = -[\rho, H]$

14. Some real life physical systems are prohibited/never occurs. This can be explained with the help of

[A] entropy [B] internal energy [C] equations of motion [D] free energy [E] none

15. The relative fluctuations in energy of a system in canonical ensemble is proportional to

[A] N [B] $1/N$ [C] N^2 [D] $1/\sqrt{N}$

16. Which of the following macrostate defines a microcanonical ensemble?

[A] $N\ V\ E$ [B] $N\ V\ T$ [C] $N\ P\ T$ [D] $P\ V\ T$

17. When a system reaches equilibrium, its entropy at equilibrium

[A] decreases [B] increases [C] is maximum [D] unpredictable

18. A three level system consists of a single particle. The particle can now be in any of these three levels. The energy of the top most level is 3ϵ, the energy of the middle level is 2ϵ and the energy of the lowest level is ϵ. The topmost energy level is five- fold degenerate, the middle one is three-fold degenerate and the lowermost one is non-degenerate. Then the canonical partition function of this system is

[A] $e^{-3\epsilon\beta} + e^{-2\epsilon\beta} + e^{-\epsilon\beta}$ [B] $e^{-15\epsilon\beta} + e^{-6\epsilon\beta} + e^{-\epsilon\beta}$

[C] $5e^{-3\epsilon\beta} + 3e^{-2\epsilon\beta} + e^{-\epsilon\beta}$ [D] $5e^{3\epsilon\beta} + 3e^{2\epsilon\beta} + e^{\epsilon\beta}$

19. The entropy associated with the arrangement of the letters in the word "PHYSICS" is

[A] $k \ln 7!$ [B] $k(\ln 7! + \ln 2!)$ [C] $k(\ln 42 + \ln 20 + \ln 3)$ [D] $k(\ln 7! - 3 \ln 2)$

20. The total number of possible microstates for a system of N coins is

[A] $2N$ [B] $N2$ [C] $2N$ [D] N [E] $N!$

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of MSc Physics Examination March 2018

PS713 Thermodynamics and Statistical Mechanics

Date: 28/03/2018 Day: Wednesday Time: 10:30AM To 01:00PM Maximum Marks: 50

Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I

Q - II Answer the following questions as directed **20**

1. Show that the work required to blow a spherical soap bubble of radius r in an isothermal, quasi-static process in the atmosphere is equal to $8\pi\gamma r^2$. Where γ is the surface tension. **2**

Soln: Two surfaces hence total area $= 2 * 4\pi r^2 = 8\pi r^2$

Work done $= 8\pi r^2 \gamma$

2. Write down the laws of thermodynamics. **3**

3. Write down the Hamiltonian of a one dimensional Harmonic oscillator. Draw the phase space diagram neatly. Obtain the Hamilton's equations of motion. **3**

Soln: $H = \frac{p_x^2}{2m} + \frac{1}{2} m\omega^2 x^2$, the phase space trajectory is an ellipse.

4. The equation of state of an ideal gas is $PV = nRT$, where *n* and *R* are constants. **4**

(a) Show that the volume expansivity β is equal to *1/T*.

(b) Show that the isothermal compressibility κ is equal to *1/P*.

Sol: $\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P = 1/T;\ \kappa = \frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T = \frac{1}{P}$

5. The equation of state of an ideal elastic substance is **4**

$\mathcal{T} = KT\left(\frac{L}{L_0} - \frac{L_0^2}{L^2}\right)$, where *K* is a constant and L_0 (the value of *L* at zero tension) is a function of temperature only. Calculate the work necessary to compress the substance from $L=L_0$ to $L=L_0/2$ quasi-statically and isothermally.

Soln: work necessary $= \int_{L_0}^{L_0/2} \mathcal{T} dl$

6. Consider a wire that undergoes an infinitesimal change from an initial equilibrium state to a final equilibrium state. Show that the change of tension is equal to **4**

$$d\mathcal{T} = -\alpha AY dT + \frac{AY}{L} dL$$

OR

Show that $\kappa_T - \kappa_S = \frac{vT\alpha_P^2}{c_P}$, where κ_T and κ_S are isothermal and adiabatic compressibility respectively, α_P is the thermal expansivity, v is the molar volume and c_P is the molar heat capacity at constant pressure.

SECTION - II

Q-III Answer the following questions as directed **30**

1 The three lowest energy levels of a certain molecule are $E_1 = 0$, $E_2 = \epsilon$, $E_3 = 10\epsilon$. Show that at sufficiently low temperature (how low?) only levels E_1 and E_2 are populated. Find the average energy E of the molecule at temperature T. **5**

Soln: $Q = 1 + e^{-\beta\epsilon} + e^{-10\beta\epsilon}$; Average energy $= -\frac{\partial \ln Q}{\partial \beta}$

OR

A two level system of $N = n_1 + n_2$ particles is distributed among two eigenstates 1 and 2with eigenenergies E_1 and E_2 respectively. The system is in contact with a heat reservoir at temperature T. If a single quantum emission into the reservoir occurs, population changes $n_2 \to n_2 - 1$ and $n_1 \to n_1 + 1$ take place in the system. For $n_1 \gg 1$ and $n_2 \gg 1$, obtain the expression for the entropy change of

(a) The two level system

(b) The reservoir

(c) From (a) and (b) derive the relation for the ratio $\frac{n_1}{n_2}$

2 Consider a large number of N localized particles in an external magnetic field H. Each particle has spin ½. Find the number of states accessible to the system as a function of M_s, the z-component of the total spin of the system. Determine the value of M_s for which the number of states is maximum. **5**

Soln: $\Omega = \frac{N!}{N_+!N_-!}$; $S = k \ln \Omega$

OR

An ideal paramagnet is placed in a magnetic field. The magnetic moments have only two possible orientations, parallel or antiparallel to the magnetic field. The Hamiltonian of the system is given by

$$H = -\sum_{i=1}^{N} m_i B$$

with $m_i = \pm m$, B as the magnetic field and N the number of magnetic moments.

(a) Calculate entropy, internal energy, magnetization, and magnetic susceptibility using the micro-canonical ensemble.

(b) Calculate all the above thermodynamic properties using canonical ensemble.

Soln: $Q = e^{-\beta H}$; $A = -kT \ln Q$, $\Omega = \frac{N!}{N_+!N_-!}$; $S = k \ln \Omega$

3 The upper end of a hanging chain is fixed while the lower end is attached to a mass M. The (massless) links of the chain are ellipses with major axes $l + a$ and minor axes $l - a$, and can place themselves only with either the major axis or the minor axis vertical. Assume that the chain has N links and is in thermal equilibrium at temperature T. **5**

(i) Find the partition function.

(ii) Find the average energy and then average length of the chain.

OR

Calculate the canonical partition function of an extreme relativistic gas consisting of N monoatomic molecules with energy-momentum relationship $\epsilon = pc$, where c is the velocity of light. Show that $PV = \frac{1}{3}U$, where U is the total internal energy of the system.

4 Consider a system of N non-interacting particles, each fixed in position and carrying a magnetic moment μ, which is immersed in a magnetic field H. Each particle may then exist in one the two energy states $E = 0$ or $E = 2\mu H$. Treat the particles as distinguishable. **5**

(a) Calculate the entropy of the system as a function of n, where n is the number of particles in the upper state.

(b) Find the value of n for which $S(n)$ is maximum.

OR

Calculate the canonical partition function of a system of N one dimensional distinguishable harmonic oscillator with similar frequency of oscillation ω. Obtain the entropy of this system.

Note: $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}$

5 A system consists of N non-interacting and distinguishable particles. Each particle can be in one of the two energy levels, $E_0 = 0$, and $E_1 = \epsilon$. The number of particles in the energy level E_1 is n_1. And the internal energy of the system is $U = n_0 E_0 + n_1 E_1$. Compute the entropy of the system and express it as a function of internal energy U of the system. Then calculate the temperature of the system. Then calculate the specific heat of the system. **5**

OR

A zipper has N links; each link has a state in which it is closed with energy 0 and a state in which it is open with energy ϵ. The zipper can only unzip from one side (say from the left) and the n th link can open only if all links to the left of it (1, 2, 3….n-1) are already open. (This model is sometimes used for DNA molecules)

(a) Find the canonical partition function.

(b) Find the average number of open links $\langle n \rangle$ and show that for low temperatures $\langle n \rangle$ is independent of N.

6 A block of copper at a pressure of 1atm (approximately 100kPa) and a temperature of 5° C is kept at constant volume. If the temperature is raised to 10° C, what will be the final pressure? The volume expansivity and isothermal compressibility values are 4.95×10^{-5} K^{-1} and 6.17×10^{-12} Pa^{-1}, respectively **5**

OR

A block of copper at a pressure of 1atm, a volume of 100 cm^3, and a temperature of 10°C experiences a rise in temperature of 5°C and an increase in volume of 0.005 cm^3. Assuming the volume expansivity and isothermal compressibility given in the above problem, calculate the final pressure.